# MOSTAKIM KHAN RAYAN

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***AI systems engineer specializing in machine learning, agentic AI, and intelligent automation across software and mechanical domains. Experienced in building scalable AI platforms, integrating autonomous pipelines, and delivering production-ready solutions from complex engineering requirements.***

Artificial intelligence–driven engineering professional with strong systems expertise across software, data, and mechanical environments. Experienced in building **machine learning** and **agentic AI workflows** for research and production applications. Skilled in developing **scalable pipelines, GPU-accelerated training systems, and AI-integrated CAD solutions**. Proven ability to **reverse-engineer complex codebases** and deploy reproducible end-to-end AI platforms. Focused on translating advanced AI capabilities into reliable, production-ready engineering systems.

## Areas of Expertise

- Machine Learning Systems
- Scalable Data Pipeline
- AI Pipeline Deployment
- Research Platform Integration
- Precision CAD Modeling
- Parametric Mechanical Design
- Computational Drug Simulation
- Scientific Data Analysis
- Systems Integration Engineering
- GPU Performance Optimization
- Reproducible Execution Frameworks
- Strategic Planning & Execution

## Career Experience

**Dr. Boran Ma's Research Lab** **November 2025 – Present**
AI Systems Integration Engineer

Architect and lead developer of an AI-powered, production-ready web platform integrating multimodal AFM analysis pipelines, unifying CNN-based classification, U-Net segmentation, and conditional Voronoi/Color Wheel modeling into a **single automated Dash application**. Designed scalable AI infrastructure to **standardize undocumented research modules**, enforce deterministic execution across independent model pipelines, and normalize heterogeneous data outputs into reproducible, research-grade results. Engineered **GPU-accelerated training** and inference environments with dynamic model loading, weight caching, fault-tolerant execution (**CUDA management, Git LFS recovery, runtime isolation**), and deployment-ready packaging to transition experimental AI code into a structured **open-source research platform**. Collaborate directly with principal investigators and researchers to align AI architecture with experimental objectives and long-term scalability for cross-institutional scientific adoption.

**Notable Achievements/Contributions:**

- Reduced AFM analysis turnaround time by ~65%, significantly increasing experimental throughput and accelerating publication timelines.
- Contributed to securing a $50,000 university research grant by delivering scalable, AI-driven analysis infrastructure that strengthened proposal competitiveness.
- Architected deployable Open-Source Platform positioned for cross-institutional adoption and AI-assisted materials research collaboration.

**Novum Dynamics LLC** **October 2025 – January 2026**
AI-Integrated Mechanical Systems Intern

Contribute to the engineering development of the proprietary Spin Dyne rotary engine by designing and refining precision parametric assemblies using **SolidWorks xDesign**. Apply AI-assisted modeling workflows, simulation-guided design iteration, and tolerance-controlled mechanical architecture to translate conceptual engine geometries into manufacturable, high-performance components. Collaborate with cross-functional engineering teams to validate rotor-casing interactions, optimize sealing geometries, and enhance structural integrity under dynamic rotational constraints. Leverage computational modeling and intelligent design tools to accelerate R&D cycles in next-generation propulsion system development.

**Notable Achievements/Contributions:**

- Advanced development of novel rotary propulsion architecture, supporting compact, high-efficiency engine innovation.
- Contributed to next-generation engine platform positioned to impact lightweight mobility, reduced mechanical complexity, and future sustainable propulsion systems.

**Vision AI Development** **May 2025 – August 2025**
Machine Learning Intern

Developed an **AI-powered fraud document detection chatbot** leveraging **PyTorch, OpenCV, and TensorFlow** to build high-precision object detection pipelines for financial document verification. Designed end-to-end **computer vision** workflows including dataset preprocessing, augmentation, model training, structured validation, and inference optimization. Integrated trained detection models into a **production-ready web application** and **scalable REST API**, enabling real-time chatbot-assisted fraud screening. Applied GPU-accelerated training strategies and performance benchmarking to ensure reliable deployment within fintech risk-control environments.

**Notable Achievements/Contributions:**

- Achieved **93–96% validation accuracy** and strong precision-recall balance in fraud document detection testing environments.
- Reduced manual document fraud review workload by approximately **50%**, accelerating verification turnaround times.
- Deployed scalable API-backed detection service supporting real-time chatbot integration for automated screening.
- Improved inference latency by **30%+** through model optimization and OpenCV-based preprocessing enhancements.

**Falcon Academy of Science** **May 2022 – August 2022**
AI Data Engineering Intern

Contributed to the development of *Bengali.ai*, a large-scale **language model** initiative, by designing and implementing structured data acquisition and preprocessing pipelines. Leveraged **API-based ingestion frameworks** to collect high-volume multilingual text datasets, applying **automated filtering, normalization, and deduplication strategies** to ensure dataset integrity. Engineered **database storage architectures** to manage structured and semi-structured training data, optimizing accessibility and scalability for downstream model training workflows. Collaborated with AI researchers to align dataset quality standards with language model development requirements.

**Notable Achievements/Contributions:**

- Curated and structured large-scale multilingual datasets to improve training data quality and linguistic coverage for Bengali language modeling.
- Reduced noise and duplication rates in collected datasets through automated filtering and normalization pipelines.
- Strengthened foundational data infrastructure for Bengali.ai language model development initiative.

## Additional Experience

**Mathematics Learning Assistant, The University of Southern Mississippi, Hattiesburg, MS,** Selected to support Pre-Calculus instruction through guided problem-solving sessions and one-on-one academic mentoring. Strengthened students' analytical reasoning, mathematical modeling, and structured problem-solving skills while collaborating with faculty to address learning gaps and improve conceptual mastery.

## Education

**Bachelor of Science in Electrical & Computer Engineering, Vanderbilt University**

**CGPA:** 4.00**, Achievements:** Robotics & Intelligence Club, The Nations Club, First place winner in Undergraduate Symposium on Research and Creative Activity (UGS) 2025 in Physics category.

## Publications

Khan, M.M. and Tuly, R.A..Examination of the underlying chemical physics of the Mpemba effect in water and other liquids.Journal of Emerging Investigators, 2023

Khan, M.M. and Tuly, R.A..A computational quantum chemical study of fluorinated Allopurinol.Journal of Emerging Investigators, 2022

## Awards

Gold Medalist, International Robotics Olympiad 2022-23

Silver Medalist, International Earth Science Olympiad 2022

Foreign Exchange Student, Asia Kakehashi Project (Japan)

Gold Awardee, The Duke of Edinburgh's International Award

## Certifications

AWS Machine Learning Foundation Badge: Credential demonstrating foundational knowledge of machine learning techniques and AWS services.

AWS Generative Artificial Intelligence Badge: Recognition of proficiency in generative AI concepts and AWS tools like Amazon Bedrock.